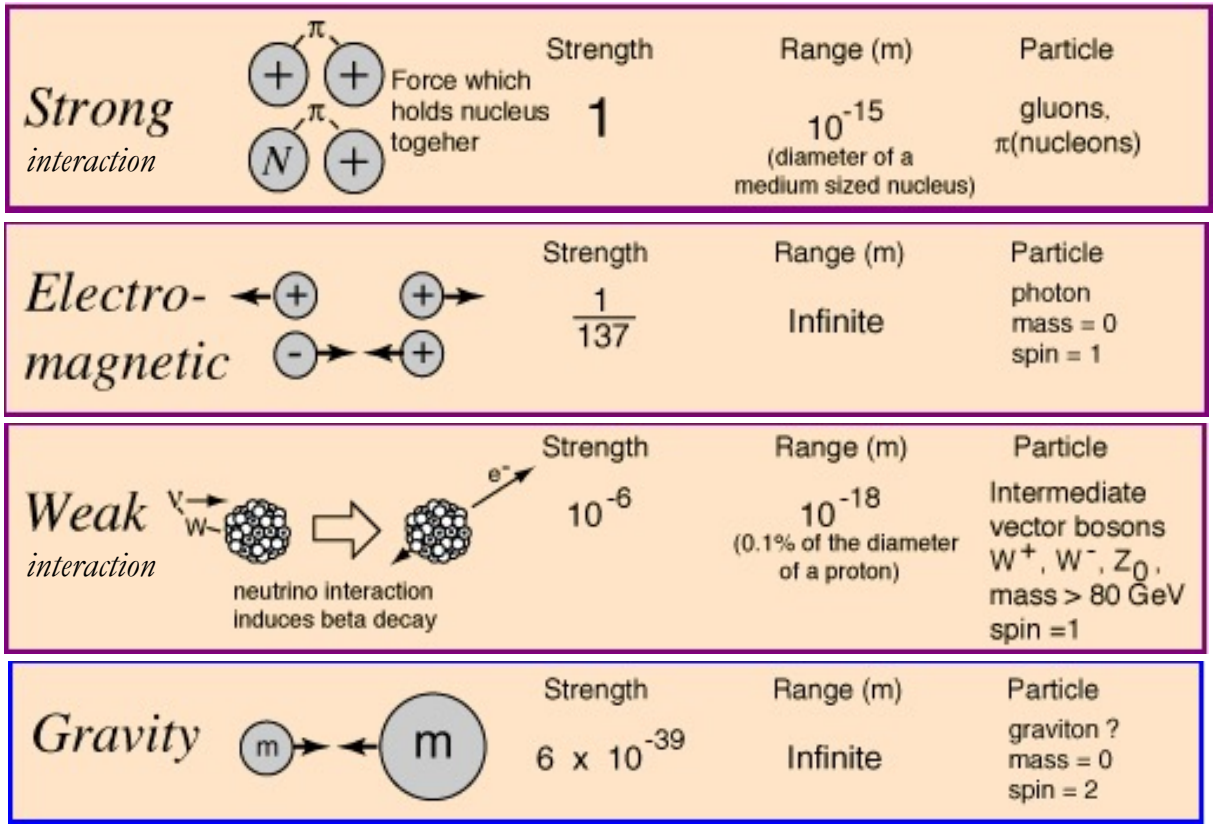
Introduction to Mechanics

What Causes Things to Change State?

The following Natural Forces are responsible for all kind of events taking place in our universe.



Motion Types

Four basic motion types. Some motions can be considered a combination of several types.

Translational: Involves change of location. A book sliding on your table.

Rotational: Involves no change of location (axis). A spinning wheel.

Oscillatory: Involves cyclic return to starting location. A pendulum in motion.

Random: Chaotic (predictable). Formation of sand domes.
Quantum (unpredictable). Electron motion at atomic scale.

Possible combinations: A bicycle in motion.

Approaches to Rigid Body Dynamics

Three different levels of treatment of rigid body dynamics will be used, depending on the situation under study.

Approach (1): Motion with constant acceleration. Equations of Motion

The force(s) causing this motion is not in the picture.
This is a simple but very limited case.

Approach (2): Motion with variable acceleration. Newton's Laws of Motion

The force(s) causing this motion are considered and evaluated.
This is a more general and practical case.

Approach (3): General dynamics. Conservation of Mechanical Energy

The different types of energies are considered and evaluated.
This is the most general and practical case.

Classical and Modern Physics

Classical Physics: Up to 1900.

- Laws of motion and gravitation.
- Conservation of energy and momentum.
- Laws of electricity and magnetism.
- Thermodynamics.

"In this field, almost everything has been discovered; all that remains is a few details."

- Based on predictability, absolute time and space.
- Consistent and agrees with reality and experiments.
- Successful: led to two industrial revolutions; the steam engine and the electric engine.

There were two issues that scientists could not satisfactorily explain:
Ether and Thermal Radiation!

Modern Physics: Starting 1900.

- Relativity: No absolute time/space, no conservation of mass/energy, c is the limit.
- Quantum Mechanics: Everything is both wave and particle, uncertainty principle.

What is Science?

Science

Richard Feynman's Philosophy of Science

Ben Trubody finds that philosophy-phobic physicist Feynman is an unacknowledged philosopher of science.



Richard Feynman (1918-88) was one of the greatest physicists of the twentieth century, contributing, among other things, to Quantum Electro Dynamics (QED), for which he won a Nobel Prize. His popular portrayal is of a buffooning genius with a preference for no-nonsense thinking – the sort that by his reckoning seemed in short supply within philosophy. He is noted, and quoted, for his dislike of philosophy, and in particular of the philosophy of science. Any quick trawl of the Internet will bring up quotes attributed to him on the absurdities of philosophy, no doubt informed by his brief flirtation with it at Princeton. Feynman would parody what he saw as 'dopey' exercises in linguistic sophistry. As he remarks in a famous lecture series, "We can't define anything precisely. If we attempt to, we get into that paralysis of thought that comes to philosophers… one saying to the other: you don't know what you are talking about! The second one says: what do you mean by 'talking'? What do you mean by 'you'? What do you mean by 'know'?" (*The Feynman Lectures on Physics*, Vol.1, 1963).

do. Putting this another way, all the description and formalizing in the world will not tell you what it is like to be a bird, or more to the point, just what it is to do science.

In 1966, Feynman addressed America's National Science Teachers Association with a talk entitled 'What is Science?' At the opening of this address, Feynman states that in textbooks on science, "There is some kind of distorted distillation and watered-down and mixed-up words of Francis Bacon" (*The Pleasure of Finding Things Out*, 2001, pp.172-3). The Baconian method was to build up universal laws from single observations via inductive reasoning – a typical example being 'This swan is white; that swan is white; and so on – from which we can draw the conclusion that all swans are white'. Feynman rebukes the Baconian concept of science, saying one cannot merely observe nature – a judgement is involved about what to pay attention to.

Here Feynman comes across as making a distinction between explicitly *knowing* things (a bit of data, say) and seeking *understanding* (a deeper and more intuitive appreciation of

Syllabus

Nano Photonics and Plasmonics Research

Instructor	Mohammad A. Alsunaidi	Office	Office Hours	Phone	Email: mohammad.alsunaidi@marmara
Course Information	**Textbook**: Fundamentals of Physics, Halliday/Resnick, 10th Ed.		Grading	Class Work 20%	Mid-term Exam 40% / Final Exam 40%

Scale document up

Week	Events	Topics to cover	Ch	Sec	Exercises
1 Oct 4-8	---	**Measurements and Units**. SI units and conversions, Dimensional analysis, Uncertainty Analysis.	1	1, 2, 3	(1) 1,9,12,13,14,21,22,27,30.
2 Oct 1-5	---	**Motion in a Straight Line**. Definitions, Instantaneous velocity, equation of motion for constant acceleration, Free-fall Analysis.	2	1, 2, 3, 4, 5	(2) 1,4,5,7,11,15,17,19,22, 23,25,27,35,37,41,44,50,52.
3 Oct 18-22	Drop/Add week	**Vector Analysis**. Vector operations, Unit vectors and decompositions, Dot and cross products.	3	1, 2, 3	(3) 2,3,9,17,24,33,36,43.
4 Oct 25-29	Cumhuriyet Bayramı Oct 29	**Motion in 2D and 3D**. Position, Displacement, Average and instantaneous velocity, Average and instantaneous acceleration, projectile motion, uniform circular motion.	4	1, 2, 3, 4, 5	(4) 3,6,7,11,14,16,22,27,28, 56,58,66,73,80.
5 Nov 1-5	Quiz 1 Nov 3	**Force and Motion I**. Newton's laws of motion, Gravitation, free-fall and weight, Applications.	5	1, 2, 3	(5) 2,4,6,10,12,13,17,21,27, 34,39,56,57.
6 Nov 8-12	---	**Force and Motion II**. Friction, Drag and terminal velocity, circular motion.	6	1, 2, 3	(6) 1,5,7,10,13,17,20,25,29, 36,39,42,44,47,57.
7 Nov 15-19	---	**Kinetic Energy and Work**. Kinetic energy, Work, Gravitational work, Work by a spring, Power.	7	1, 2, 3, 4, 5, 6	(7) 1,5,7,11,14,15,16,19,23, 27,29,34,35,36,41,43,46,50.
Nov 20-28	**Mid-term Exams period**				
8 Nov 29-Dec 3	---	**Conservation of Energy**. Potential energy, Conservation of mechanical energy, Work by external forces, Conservation of energy.	8	1, 2, 3, 4, 5	(8) 1,2,5,7,9,13,15,16,23,29, 42,48,49,53,57.
9 Dec 6-10	---	**Center of Mass and Linear Momentum**. Center of mass, System of particles, Linear momentum.	9	1, 2, 3	(9) 1,2,3,4,9,10,14,18,20,25.
10 Dec 13-17	---	**Center of Mass and Linear Momentum**. Collision and impulse, Conservation of linear momentum, momentum and kinetic energy, Elastic collisions.	9	4, 5, 6, 7	(9) 27,32,35,38,46,49,50,52.
11 Dec 20-24	---	**Rotation**. Definitions, Rotation with constant angular acceleration, linear and angular variables, kinetic energy of rotation, Inertia, Torque, Newton's laws for rotation, Work.	10	1-8	(10) 2,4,7,9,10,11,14,19, 22, 23,34,36,46,48,50,51,58,60.
12 Dec 27-31	New Year 2022	**Angular Momentum**. Torque, Angular momentum, Newton's second law in angular form.	11	4, 5, 6	(11) 19,21,25,26,30,32, 35,36,37.
13 Jan 3-7	Quiz 2 Due Jan 5	**Angular Momentum**. Angular momentum of a rigid body, Conservation of angular momentum.	11	7, 8	(11) 40,49,50,59,60.
14 Jan 10-14	---	**Static Equilibrium and Elasticity**. Static equilibrium, Examples, Elasticity.	12	1, 2, 3	(12) 2,5,10,15,21,23,31,33, 38,43,44,48.
15 Jan 17-21	---	**Review**	---	---	---
Jan 24- Feb 6	**Final Exams period**	**Feb 8-14 (Re-sit Final Exams period)**			

Chapter 1: Measurements and Units

The following subjects will be discussed

- SI Units.
- Unit analysis and conversion.
- Scientific notation.
- Dimensional analysis.
- Uncertainty.

Physics and Measurements

Work in physics is based on measurements.

In order to study a physical phenomenon, we design an experiment.

An experiment is a method of interrogating nature. The results we get from experiment are nature's answers.

We need to understand these answers, by:

1) Asking the right questions (experiment design).

2) Pay attention to answers (equipment selection).

3) Interpret results (analysis).

4) Finding patterns and laws (mathematical frame).

We need Units.

Metrology

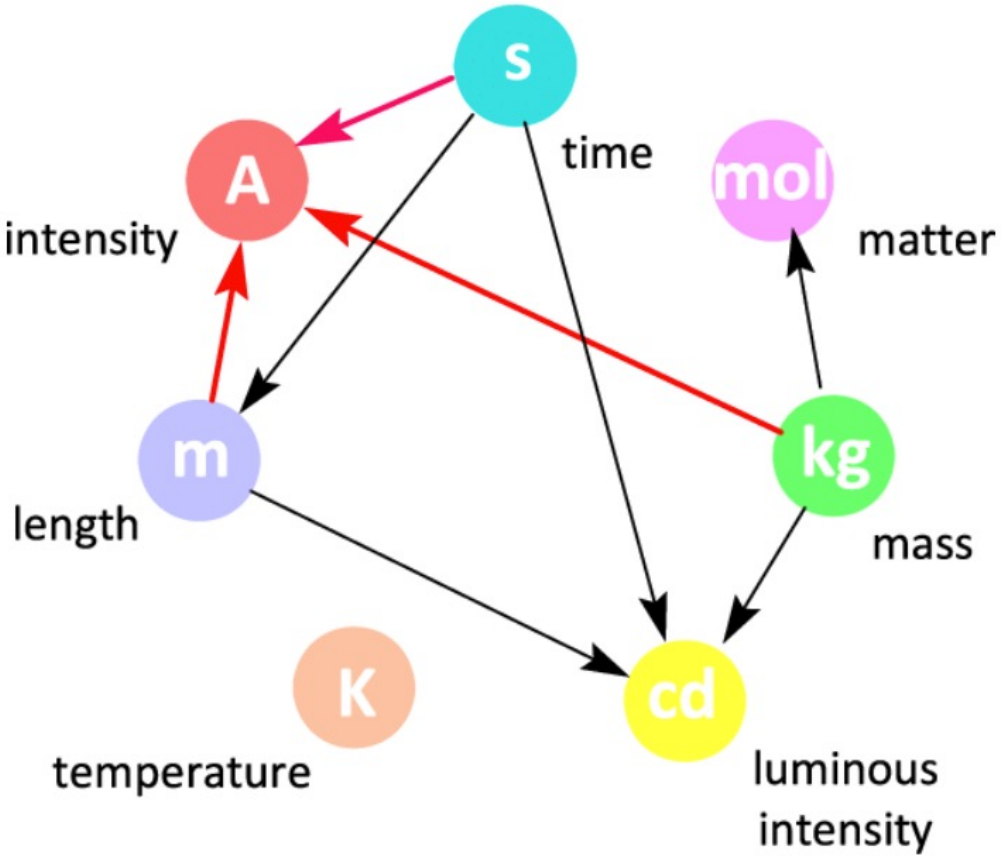
The written history of Metrology dates back to the Egyptians. Tools and scriptures indicate they have used a number of length and weight measurements.



SI Units

The International System of units (SI units) is the modern form of metric system which is officially recognized in almost all labs around the world.

The definitions of these units have gone through a lot of adjustments over the years. The most up-to-date definitions use the well-known universal quantities as references.



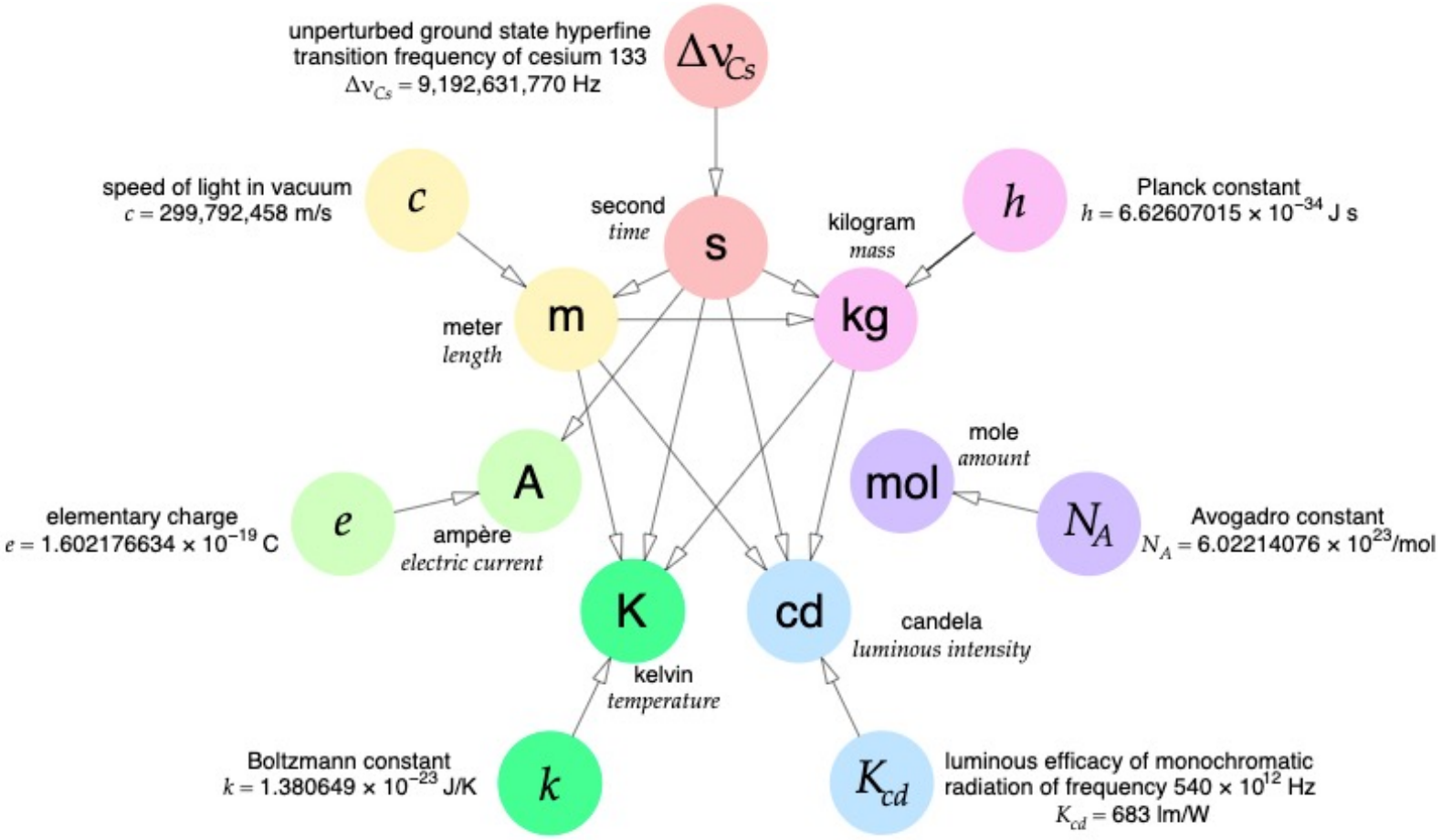
Universal and Cultural Constants

It would be great if the basic units were tied to some
Universal Constants;
constants that come from the fabric of the universe.

symbol	value	description	
$\Delta\nu_{Cs}$	9,192,631,770 Hz	the unperturbed ground state hyperfine transition frequency of the cesium 133 atom	
c	299,792,458 m/s	the speed of light in vacuum	Universal Constant
h	$6.62607015 \times 10^{-34}$ J s	Planck's constant	Universal Constant
e	$1.602176634 \times 10^{-19}$ C	the elementary charge	Universal Constant
k	1.380649×10^{-23} J/K	Boltzmann's constant	Universal Constant
N_A	$6.02214076 \times 10^{23}$ 1/mol	Avogadro's constant	
K_{cd}	683 lm/W	the luminous efficacy of monochromatic radiation of frequency 540×10^{12} Hz	

SI Units

Example: the second is defined by taking the fixed numerical value of the Cs frequency $\Delta\nu_{Cs}$, the unperturbed ground-state hyperfine transition frequency of the ^{133}Cs atom, to be 9192631770 when expressed in the unit Hz, which is 1/s.



Units Analysis and Conversion

In many instances we would like to find out the basic units of some quantities, like force, energy, etc.

Example: the units of Newtonian force.

$$\vec{F} = m\vec{a} \qquad \vec{a} = \frac{d\vec{v}}{dt} \qquad \vec{v} = \frac{d\vec{x}}{dt}$$

$$N = kg \cdot m/s^2 \qquad m/s^2 \qquad m/s$$

Example: the units of Power.

$$P = \frac{dU}{dt} \qquad U = \frac{1}{2}mv^2 \qquad v = \frac{dx}{dt}$$

$$W = \frac{J}{s} = kg \cdot m^2/s^3 \qquad kg \cdot m^2/s^2 \qquad m/s$$

Units Analysis and Conversion

In many instances the physical quantities are not expressed in the basic units; unit conversion is needed.

Example: A motorist is driving at 0.02 mi/s on the E5, will he be cited?

80 km/h

Speed Limits on the **E5** Highway :

Cars: 80 km/h. Buses and trucks: 80 km/h. Motorbikes: 70-80 km/h.

$$\text{Speed} = 0.02 \frac{mi}{s} \cdot \frac{1.61\ km}{1\ mi} \cdot \frac{3600\ s}{1\ hr} = 116\ \frac{km}{hr}$$

Scientific Notation

The scientific notation unifies the expression of numbers (especially too large or too small). It is also easy to visualize.

Electron charge

$$1.60217\times10^{-19}\ \text{C}$$

Speed of light

$$3.0\times10^{8}\ \text{m/s}$$

Order of Magnitude allows us to categorize or compare two quantities.

Mosquito mass $1\times10^{-5}\ kg$

Human mass $1\times10^{2}\ kg$

Humans are 7 orders of magnitude more massy.

DECIMAL MULTIPLIER PREFIXES

Prefix	Symbol	Multiplier
exa	E	10^{18}
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
hecto	h	10^{2}
deka	da	10^{1}
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}

Uncertainty Analysis

How do we record measurements in experiments and how do we process them afterwards? How to deal with uncertainty in measurements and calculations? There are a few rules.

Example: Let's say we have done measurements on two quantities and the results are

$$a = 3.00 \pm 0.03 \text{ (m)} \qquad b = 1.00 \pm 0.02 \text{ (m)}$$

Uncertainty within 3 cm. Uncertainty within 2 cm.

$$\Delta a = 0.03 \text{ (m)} \qquad \Delta b = 0.02 \text{ (m)}$$

1) How much is the uncertainty in $S = a + b$ and $S = a - b$?

For addition and subtraction, add uncertainties.

$$S = a + b = 4.00 \text{ (m)}$$

$$\Delta S = \Delta a + \Delta b = 0.05 \text{ (m)}$$

$$S = 4.00 \pm 0.05 \text{ (m)}$$

$$S = a - b = 2.00 \text{ (m)}$$

$$\Delta S = \Delta a + \Delta b = 0.05 \text{ (m)}$$

$$S = 2.00 \pm 0.05 \text{ (m)}$$

Q & A

Send me an email if
you have any
questions.